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Recommendations for Comparison of Productivity Between Fed-Batch and Perfusion Processes

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Due to the growing interest in integrated continuous processing in the biopharmaceutical industry, productivity comparison of batch-based and continuous processes is considered a challenge. Integrated continuous manufacturing of biopharmaceuticals requires scientists and engineers to collaborate effectively. Differing definitions, for example, of volumetric productivity, may cause confusion in this interdisciplinary field. Therefore, the aim of this communication is to reiterate the standard definitions and their underlying assumptions. Applying them to an exemplary model scenario allows to demonstrate the differences and to develop recommendations for the comparison of productivity of different upstream processes.

1. Introduction

Biopharmaceutical processing is an interdisciplinary field, which makes it both challenging and interesting. In the early days, especially mammalian cell culture processing was dominated by biologists and immunologists. Successful bioreactor scale-up required chemical engineering knowledge. With the current trend to integrate upstream (cultivation of organisms producing the protein of interest) and downstream (product purification) processing to next generation bioprocesses,[1,2] it is important to align on terminology. For example, different descriptions for product concentration (e.g., "titer") and volumetric productivity of upstream processes are in use.[3] This paper reviews the standard productivity definitions, and gives guidance on how it should be calculated and compared for different upstream processes (batch or continuous).

2. Experimental Section

Characteristic for perfusion processes is a continuous "feed" of fresh medium in the bioreactor (perfusion rate; *P*) combined with a continuous outflow of cell free harvest (harvest rate; *H*). Perfusion and harvest rate need to be equal to achieve a constant culture volume. To achieve and maintain a steady-state, a "cell bleed" (*B*) needs to be discarded from the bioreactor, increasing the perfusion rate.

$$P = B + H \tag{1}$$

Equation (1): Definition of perfusion rate. P = perfusion rate [d^{-1}], B = bleed rate [d^{-1}], H = harvest rate [d^{-1}].[4]

Flow rates are conventionally expressed in specific rates, that is, volume of medium per bioreactor working volume per day (vvd; [d^{-1}]).[4] In case the cell retention device (CRD) does not allow 100% cell retention, the harvest rate can be mathematically split into a cell-free harvest and a bleed stream equal to the bioreactor cell density that can be added to the bleed rate.[3]

Key variables of perfusion processes are depicted in **Figure 1**. To characterize the cellular medium requirement, cell specific perfusion rate (CSPR) is typically considered.[4,5] CSPR indicates nutrient supply per cell and day in a continuous process and is cell line and medium specific.

Based on the perfusion rate and the viable cell density (VCD), the CSPR can be calculated as:

$$\mathrm{CSPR} = \frac{P}{X} \tag{2}$$

Equation (2): CSPR determination. CSPR = Cell specific perfusion rate [$nL\,cell^{-1}\,d^{-1}$], P = perfusion rate [d^{-1}], X = cell density [10^6 cells mL^{-1}].[4]

In contrast to perfusion rate, cell specific perfusion rate (CSPR) is media consumption normalized to cell density, and therefore constitutes an important perfusion medium performance criterion.[6] A constant CSPR for perfusion rate control provides an optimum supply of medium per cell, for example, in exponential growth phase.[5,7] The minimum CSPR (CSPR_{min}) is defined as the minimum perfusion rate that is required for the cultivation of a specific cell line without suffering from a nutrient limitation. Below this value, growth rate, viability, or specific productivity may decrease. CSPR values that are close to CSPR_{min} allow product concentrations in the harvest stream near fed-batch processes, decrease medium demand, and increase medium utilization.[4] On the other hand, if steady-state processes are targeted, a certain offset ("safety margin," *S*) must be added to CSPR_{min} to avoid nutrient limitations through temporary medium supply interruption or process variability, resulting in CSPR_{crit}.[3]

$$\mathrm{CSPR}_{crit} = (1 + S) * \mathrm{CSPR}_{min} \tag{3}$$

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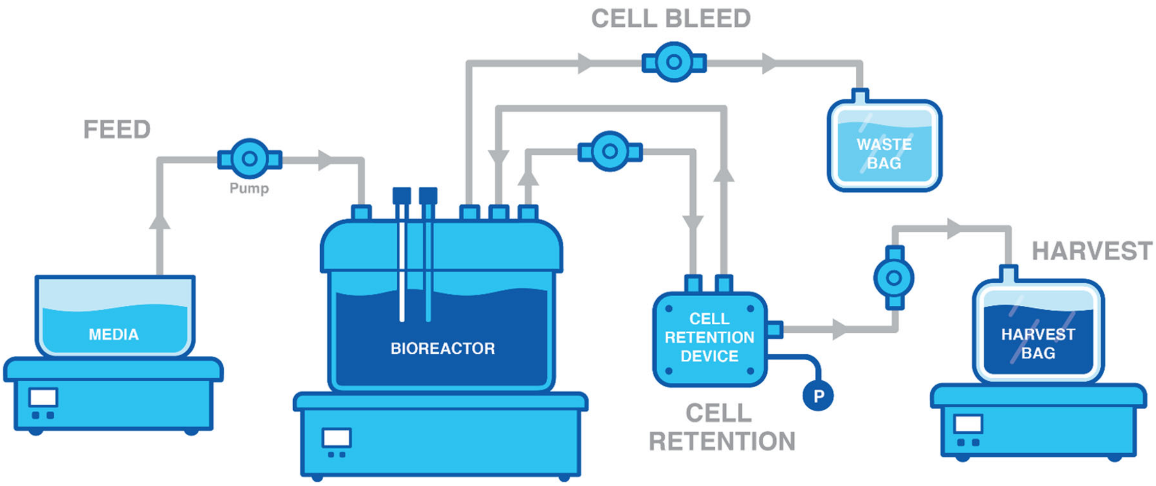

Figure 1. Typical steady-state perfusion bioreactor setup.

Equation (3): Definition of $\mathrm{CSPR_{crit}}$ and safety margin. CSPR = Cell specific perfusion rate [nL cell^{-1} d^{-1}] P = perfusion rate [d^{-1}], S = offset constant or "safety margin" [–].

There are various possibilities of comparing productivity of fed-batch and perfusion processes, for example, yield, volumetric productivity, or space-time yield (STY).

Yield is typically defined as the mass of accumulated product since process start.[3] For fed-batch processes, it is equal to the product concentration in the bioreactor at a specified day i, multiplied by the current bioreactor working volume:

$$Y_{\mathrm{Fed-Batch},i} = c_{P,i} * V_{\mathrm{Bioreactor},\,i} \tag{4}$$

Equation (4): Yield. Y = yield [g], c_P = product concentration [g L^{-1}], V = working volume [L].

As the biomass in high density and "concentrated" fed-batch cultures takes up a significant part of the culture volume, the effective liquid volume can be significantly lower than the bioreactor working volume.[8]

For perfusion processes, the harvest flow rate and current product concentration need to be considered. The daily yield equals the accumulated mass of product since process start:

$$Y_{\mathrm{Per\ fusion},\,i} = \int_i^0 c_{P,i} * H_i * V_{\mathrm{Bioreactor}} \tag{5}$$

Equation (5): Yield. Y = yield [g], c_P = product concentration [g L^{-1}], H = harvest rate [d^{-1}], V = working volume [L].

For comparison of the overall productivity of upstream processes, space-time yield (STY) can be used which should not be confused with volumetric productivity (VP). It equals the yield Y divided by the process duration and culture volume.

$$\mathrm{STY}_i = \frac{Y_i}{V_{\mathrm{Bioreactor}} * (t_i - t_0)} \tag{6}$$

Equation (6): Space-time yield. STY [g L^{-1} d^{-1}], Y = yield [g], t = process time [d], V = working volume [L] (adapted from ref. [9,10]).

For Fed-Batch processes, Equation (6) can be simplified to:

$$\mathrm{STY}_{\mathrm{Fed-Batch},\,i} = \frac{c_{P,i}}{(t_i - t_0)} \tag{7}$$

Equation (7): Space-time yield for Fed-Batch processes. STY [g L^{-1} d^{-1}], c_P = product concentration [g L^{-1}], t = process time [d].

While STY describes the overall productivity of a process, volumetric productivity (VP [g L^{-1} d^{-1}]) typically refers to the current productivity at a given time. For Fed-Batch processes, VP can be determined according to:

$$\mathrm{VP}_{\mathrm{Fed-Batch},\,i} = \frac{dc_P}{dt} = \frac{c_{p,i} - c_{p,i-1}}{t_i - t_{i-1}} \tag{8}$$

Equation (8): Volumetric productivity. VP = volumetric productivity [g L^{-1} d^{-1}], $c_{P,i}$ = product concentration on day i [g L^{-1}].

For VP in perfusion, the current product concentration is multiplied with the harvest rate and describes the current productivity at a given time:

$$\mathrm{VP}_{\mathrm{Per\ fusion},\,i} = c_{P,i} * H_i = X_i * q_{P,i} \tag{9}$$

Equation (9): Volumetric productivity for perfusion processes. VP = volumetric productivity [g L^{-1} d^{-1}], $c_{P,i}$ = concentration of product on day i [g L^{-1}], H = harvest rate [d^{-1}], X = cell density [10^6 cells mL^{-1}], q_P = cell specific productivity [pg cell^{-1} d^{-1}].[4]

In steady-state perfusion processes, the product is usually harvested continuously from the bioreactor, which leads to lower product concentrations compared to fed-batch processes.

For the development and optimization of cell culture media specifically for perfusion applications, the specific production rate q_P is a critical target.[11,12] Specific productivity per cell can be calculated from medium exchange rate considering constant flow rates, viable cell density and the product concentration in the supernatant.

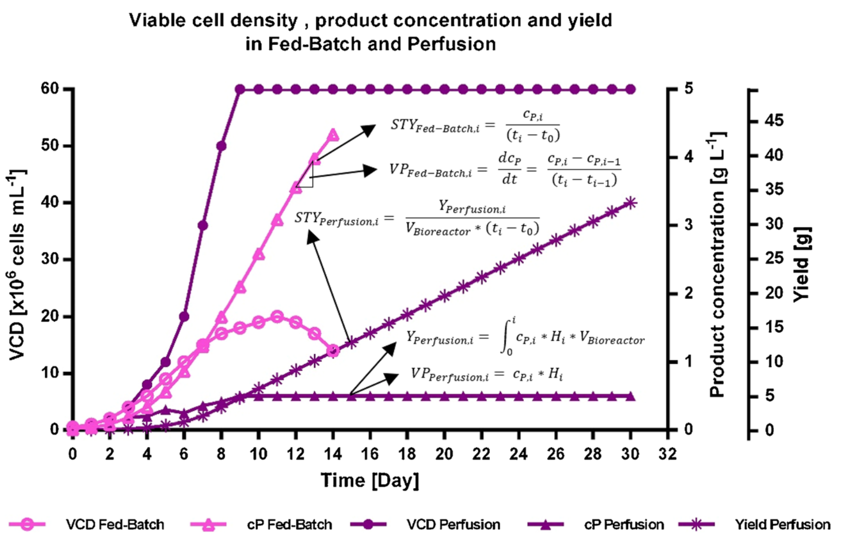

Figure 2. Exemplary model scenario of Fed-Batch and Perfusion processes. Graph illustrates viable cell density (VCD), product concentration (c_P) and yield (Y). Equations indicate how space-time yield (STY) and volumetric productivity (VP) in **Figure 3** are calculated.

$$\frac{dc_p * V}{dt} = F_{\text{in}} * c_{p,\text{in}} - F_{\text{out}} * c_{p,\text{out}} + r * V$$

$$F_{\text{in}} = F_{\text{out}} = F$$

$$P = \frac{F}{V}$$

$$\frac{\mathrm{d}c_p}{\mathrm{d}t} = P * (c_{p,\text{in}} - c_{p,\text{out}}) + r$$

$$r = q_p * X$$

$$\frac{\mathrm{d}c_p}{\mathrm{d}t} = \mathrm{P} * (c_{p,\text{in}} - c_{p,\text{out}}) + q_p * X \quad (10)$$

Equation (10): Determination of metabolic rates in perfusion. V = working volume [L], F = flow [L d^{-1}], P = perfusion rate [d^{-1}], c_P = substrate concentration [g L^{-1}], r = reaction rate [g L^{-1} d^{-1}], q_P = specific productivity [pg cell^{-1} d^{-1}], X = cell density [10^6 cells mL^{-1}].[13]

The metabolic rates of substrate consumption or metabolite production can be determined the same way. If the determination of cellular productivity (q_P) in perfusion, Equation (11) can be used. To avoid an underestimation of q_P, the perfusion rate needs to be used for the determination instead of the harvest rate as product can be lost through the application of a bleed strategy.

$$q_{P,i} = \frac{(c_{P,i} - c_{P,i-1})}{(t_i - t_{i-1})} + P * c_{P,i} * X^{-1} \quad (11)$$

Equation (11): Cellular productivity in perfusion. q_P = specific cellular productivity [pg cell^{-1} d^{-1}], P = perfusion rate [d^{-1}], c_P = product concentration [g L^{-1}], X = cell density [10^6 cells mL^{-1}] .

3. Results

In this section, we will present an example to illustrate the theory discussed above. For this, we will use data from a case study in which a CHO cell line producing a monoclonal antibody was cultivated both in fed-batch and perfusion. Both reactors are inoculated at 0.5*10^6 cells mL^{-1}. The fed-batch process has a 14 days duration, a maximum VCD of 20*10^6 cells mL^{-1}, and a final product concentration of 4.3 g L^{-1}. The perfusion process is first operated in batch mode, then at increasing perfusion rates between 1 and 3 vvd. VCD reached 60*10^6 cells mL^{-1} on day 10 after inoculation, when a constant bleed of 0.3 vvd was used to establish steady-state. The process is reported for a duration of 30 days. Some assumptions were made to make the comparison between the process types clear: The example cell line used in both processes has a specific productivity of 25 pg cell^{-1} d^{-1}, which is constant for all growth phases and process types. This results in a steady-state product concentration of 0.5 g L^{-1} in perfusion. Specific productivity can be influenced by many variables, for example, process, medium, cell specific perfusion rate (CSPR), temperature, pH etc.[3] which are not considered here. Also, in real processes, fluctuations in growth rate and, consequently, bleed rate can occur. Cell growth, product concentration, and yield time course are depicted in **Figure 2**, along with the equations of important productivity indicators which can be calculated from these.

Volumetric productivity (VP) and space-time yield (STY) for these processes are presented in Figure 3. For perfusion processes, it must be considered that the volume used to maintain steady-state by bleeding is typically considered lost and only the harvest stream is considered to calculate yield. In this example, titer is 0.5 g L^{-1}, and P is 3 vvd, but due to a bleed rate B of 0.3 vvd, harvest rate H is only 2.7 vvd, resulting in an effective VP of 1.35 g L^{-1} d^{-1} in steady-state. In fed-batch, VP is changing from day to day, depending on the increase of product concentration (Figure 3). The maximum VP of 0.5 g L^{-1} d^{-1} is on day 11, coinciding with peak VCD in consequence of the assumption of constant q_P. This suggests that the steady-state VP of perfusion is 2.7-fold higher than the maximum VP of the corresponding fed-batch process, stemming from a three-fold increase in cell density using perfusion technology.

For all following calculations, a reactor size of 1 L is considered. For the fed-batch, this is the final culture volume including feeds and base additions. Yield in this example is 4.34 g for fed-batch on day 14 and 33.04 g for the perfusion process on day 30. From the daily yield values and process duration, STY can be calculated (Figure 3). The example data suggests an STY of 0.31 g L^{-1} d^{-1} for fed-batch on day 14 and of 1.10 g L^{-1} d^{-1} for perfusion on day 30, a 3.5-fold increase. To

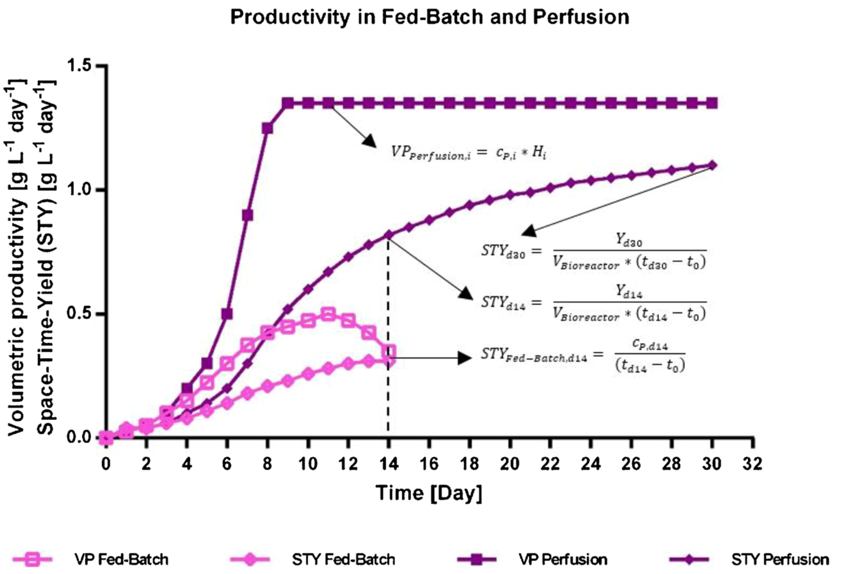

Figure 3. Exemplary model scenario of Fed-Batch and Perfusion processes. Graph illustrates volumetric productivity (VP) and space-time yield (STY) calculated from modeled values in **Figure 2**.

compare equal bioreactor operation times, on day 14, due to the higher VCD, STY in perfusion is 0.82 g L^{-1} d^{-1}, still a more than 2.7-fold increase over fed-batch processing.

One could argue that, within a 30-day time slot, two 14 day fed-batch processes could be performed considering a 1-day turnaround time. Performing two fed-batch processes would obviously double the yield. However, in terms of productivity, STY would first decrease to 0.23 g L^{-1} d^{-1} on day 20, and then increase again to 0.29 g L^{-1} d^{-1} on day 30 (Figure S1, Supporting Information). The bioreactor turnaround of 1 day, that was not considered in the first fed-batch process, has decreased STY by 0.02 g L^{-1} d^{-1}. In a regulated manufacturing environment, turnaround times may be significantly longer, even if single-use equipment has helped to reduce these throughout the biopharmaceutical industry. Effective STY, that is, including turnaround times, will be lower than theoretical STY, thereby decreasing plant capacity. Perfusion, on the contrary, provides a constant STY in steady-state. Longer process times in perfusion also decrease the impact of bioreactor turnaround on effective productivity.

4. Discussion

Different possibilities of determining and comparing the productivity of upstream processes have been presented. Volumetric productivity refers to the current productivity and the question is, which time point should be used for comparison. We recommend a comparison of the most productive day of a fed-batch process with the steady-state productivity in perfusion. For space-time yield, comparison of different time points can be considered. It must be noted that the productivity of a manufacturing plant will depend also on bioreactor turnaround times, scheduling, and other potential bottlenecks, which decrease the effective productivity below the theoretically possible space-time yield.

Abbreviations

CHO, Chinese hamster ovary (cell line); CRD, cell retention device; CSPR, cell specific perfusion rate; STY, space-time yield; VCD, viable cell density; VP volumetric productivity; vvd, volume per volume per day.

Conflict of Interest

The authors are employees of Merck KGaA, Darmstadt, Germany, and declare no financial or commercial conflict of interest.

Keywords

CHO, perfusion, productivity comparison, space-time yield, upstream processes

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